

Programming Using C++ Level 1

Arithmetic Operators



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Arithmetic Operators

Definition

◆ Arithmetic operators are symbols used to perform mathematical calculations in a program.

□ The main arithmetic operators in C++ are:

`+` → Addition

`-` → Subtraction

`\*` → Multiplication

`/` → Division

`%` → Modulus (remainder)

Explanation

◆ In this example, the program asks the user to enter three marks and then calculates their average.

```
int Mark1;
```

```
int Mark2;
```

```
int Mark3;
```

- Mark1 → stores the first mark.

- Mark2 → stores the second mark.

- Mark3 → stores the third mark.

◆ The program calculates the average using:

```
int Average = (Mark1 + Mark2 + Mark3) / 3;
```

- + → adds the three marks.

- / → divides the total by 3.

- () → ensures that the addition is performed before the division.

□ Steps in main()

1 Declare Mark1

◆ Create a variable to store the first mark.

2 Ask for the first Mark

◆ Use cin to read the first mark from the user.

3 Declare Mark2

◆ Create a variable to store the second mark.

4 Ask for the second Mark

◆ Use cin to read the second mark from the user.

5 Declare Mark3

◆ Create a variable to store the third mark.

6 Ask for the third Mark

◆ Use cin to read the third mark from the user.

7 Calculate the Average

◆ Add the three marks and divide the result by 3.

8 Display the Average

◆ Print the calculated average to the screen.

9 End the program

◆ Use return 0; to finish the program successfully.

Example

```
int Average = (Mark1 + Mark2 + Mark3) / 3;
```

◆ If the user enters:

Mark1 = 80

Mark2 = 90

Mark3 = 70

◆ The calculation is:

$(80 + 90 + 70) / 3$

$240 / 3$

80

→ Results

◆ The program displays:

Average = 80

Note

- ◆ The arithmetic operators used in this example are:
 - $+$ $\rightarrow$ Addition
 - $/$ $\rightarrow$ Division
- ◆ Because Mark1, Mark2, and Mark3 are int variables, the result of the division is also treated as an integer.

⊘ Warning

- ◆ When dividing two integers, C++ performs integer division.
- ◆ For example:
 `int Average = 10 / 3;`
- ◆ The result will be:
 3
- ◆ The decimal part is discarded.



Comparison

□ Addition +

- ✓ Adds values together.
- ✓ Example: $10 + 5$
- ✓ Result: 15

□ Division /

- ✓ Divides one value by another.
- ✓ Example: $10 / 5$
- ✓ Result: 2

Summary

- ✓ Arithmetic operators are used to perform mathematical calculations.
 - $+$ → Addition
 - $-$ → Subtraction
 - $*$ → Multiplication
 - $/$ → Division
 - $\%$ → Remainder
 - $()$ → Controls the order of calculations.

Thank you 😊